Graph Theory II: Trees and Minimum Spanning Trees

Walks and Paths

We begin with the fundamental notions of traversal in a graph.

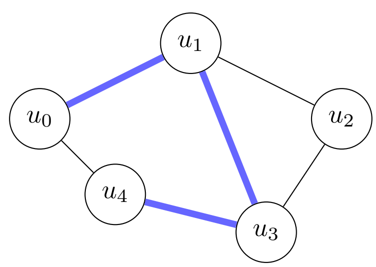
Definition. A **walk** in a graph $G = (V, E)$ is a sequence of vertices

$$u_0,\ u_1,\ \ldots,\ u_k$$

such that $\{u_{i-1}, u_i\} \in E$ for each $i = 1, \ldots, k$. The *length* of the walk is k, the number of edges traversed.

Definition. A **path** is a walk $u_0, u_1, \ldots, u_k$ in which all vertices are distinct.

Any walk from u_0 to u_k contains a path from u_0 to u_k: whenever a vertex is revisited, the segment between the two occurrences forms a cycle that can be short-circuited. Repeating this step yields a path.

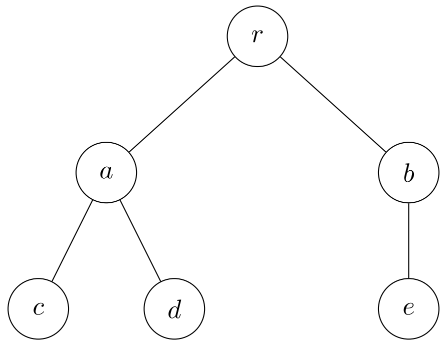


Thick blue edges: path $u_0 \to u_1 \to u_3 \to u_4$ inside the graph.

★ **EXAM PRIORITY.** Know the distinction between a **walk** and a **path**. A walk may revisit vertices and edges; a path visits each vertex at most once. Be able to identify or extract a path from a given walk.

Trees

Definition. A **tree** is a connected, acyclic graph. A **forest** is an acyclic graph whose connected components are trees.



A tree on 6 vertices and $5 = 6 - 1$ edges.

Theorem 1. *Every tree with $n \geq 1$ vertices has exactly $n - 1$ edges.*

Proof. By induction on n.

Base case ($n = 1$). A single vertex has no edges, and $1 - 1 = 0$.

Inductive step. Assume every tree on n vertices has $n - 1$ edges. Let T be a tree on $n + 1$ vertices.

Since T is connected and acyclic with at least two vertices, it has at least one *leaf* (a vertex of degree 1). Let v be a leaf of T, and let T' be the subgraph obtained by deleting v and its unique incident edge. We claim T' is a tree:

- *Acyclic.* T' is a subgraph of the acyclic graph T.
- *Connected.* Any path in T between two vertices of T' cannot pass through v: since v has degree 1, a walk entering v must leave by the same edge, violating the path property. Hence all pairwise connections survive in T'.

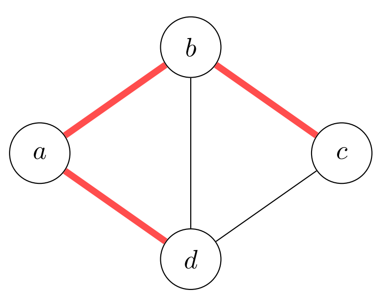
T' is a tree on n vertices, so by the inductive hypothesis it has $n - 1$ edges. Restoring v adds exactly one edge, giving T a total of $(n - 1) + 1 = n$ edges. $\square$

★ **EXAM PRIORITY. Key fact.** A tree on n vertices has exactly $n - 1$ edges. Know this, and be prepared to prove it by induction using the leaf-deletion argument above.

Spanning Trees

Definition. A **spanning tree** of a connected graph $G = (V, E)$ is a subgraph $T = (V, F)$ with $F \subseteq E$ that is a tree (connected, acyclic, on all $|V|$ vertices).

Every connected graph has at least one spanning tree: remove edges from cycles one at a time until none remain.



All edges form G. Red (thick) edges form one spanning tree (4 vertices, $3 = 4{-}1$ edges).

Minimum Spanning Trees

Definition. Given a connected weighted graph $G = (V, E)$ with weight function $w : E \to \mathbb{R}$, a **minimum spanning tree** (MST) is a spanning tree $T = (V, F)$ minimizing $w(T) = \sum_{e \in F} w(e)$.

Edge-Swap Lemma

The following structural lemma underlies every efficient MST algorithm.

Theorem 2 (Edge-Swap)**.** *Let $T^* = (V, E^*)$ be an MST of G, and let $e \notin E^*$. Adding e to T^* creates exactly one cycle C. For every $e' \in C \setminus \{e\}$,*

$$w(e') \le w(e).$$

Proof. Because T^* is a spanning tree it is connected, so adding $e = \{u, v\}$ introduces exactly one cycle: the unique u-v path in T^* together with e.

Suppose for contradiction some $e' \in C \setminus \{e\}$ satisfies $w(e') > w(e)$. Define

$$T^{**} = \big(V,\ (E^* \setminus \{e'\}) \cup \{e\}\big).$$

Removing e' splits T^* into two components; re-adding e (which lies on the same cycle C) reconnects them, so T^{**} is connected. T^{**} is acyclic because e' was the only edge removed from the unique cycle C before e was inserted. Hence T^{**} is a spanning tree, and

$$w(T^{**}) = w(T^*) - w(e') + w(e) < w(T^*),$$

contradicting the minimality of T^*. Therefore $w(e') \le w(e)$. $\square$

Kruskal's Algorithm and the Greedy Invariant

Definition. Kruskal's Algorithm. Initialize $S \leftarrow \emptyset$. Repeatedly select the minimum-weight edge $e \in E \setminus S$ such that $S \cup \{e\}$ is acyclic, and set $S \leftarrow S \cup \{e\}$. Terminate when $|S| = |V| - 1$.

Theorem 3 (Greedy Invariant)**.** *Let S_m denote the first m edges chosen by Kruskal's Algorithm. There exists an MST $T^* = (V, E^*)$ with $S_m \subseteq E^*$.*

Proof. By induction on m.

Base case ($m = 0$). $S_0 = \emptyset \subseteq E^*$ for any MST T^*.

Inductive step. Assume $S_{m-1} \subseteq E^*$ for some MST $T^* = (V, E^*)$. Let e be the m-th edge chosen. If $e \in E^*$ we are done immediately.

Suppose $e \notin E^*$. Adding $e = \{u, v\}$ to T^* creates a cycle C. We claim some $e' \in C \setminus \{e\}$ satisfies $e' \notin S_{m-1}$: if every edge of $C \setminus \{e\}$ were in S_{m-1}, then S_{m-1} would contain a u-v path, making $S_{m-1} \cup \{e\}$ cyclic—contradicting that the algorithm accepted e. Choose such e'.

Since $S_{m-1} \cup \{e'\} \subseteq E^*$ and E^* is acyclic, e' is a valid acyclic extension of S_{m-1}. The algorithm selected the minimum-weight such extension, so $w(e) \le w(e')$. The Edge-Swap Lemma gives $w(e') \le w(e)$; hence $w(e) = w(e')$. Define

$$T^{**} = \big(V,\ (E^* \setminus \{e'\}) \cup \{e\}\big).$$

By the same connectivity and acyclicity argument, T^{**} is a spanning tree with $w(T^{**}) = w(T^*)$, hence an MST. Since $e' \notin S_{m-1}$,

$$S_m = S_{m-1} \cup \{e\} \ \subseteq\ (E^* \setminus \{e'\}) \cup \{e\} = E^{**}.$$

The invariant holds at step m. $\square$

When Kruskal's Algorithm terminates it has produced a spanning tree T_{out} on $|V| - 1$ edges. By the Greedy Invariant, $T_{\text{out}} \subseteq E^*$ for some MST T^*; since both are spanning trees of the same size, $T_{\text{out}} = T^*$ is itself an MST.

★ **EXAM PRIORITY. Greedy MST (Kruskal's).** At each step add the cheapest edge that does not form a cycle; stop after $n - 1$ edges. Correctness rests on two pillars: (1) every spanning tree on n vertices has $n - 1$ edges, and (2) the Edge-Swap Lemma—at every step the algorithm's edge set is a subset of some MST.
